

## Assignment #3

Section A: Answer the following Questions:

(5x2=10 marks)

1. Explain the constructor method. How it differs from other member function.
2. Differentiate between default and parameterized constructor?
3. Why is constructor used under public access specifier?
4. Complete below code:

```
Class Student{
    int roll_number ;
    string name ;
    public:
        //initialize above data members using initializer list
        Student(){
            }
        };
    int main(){
        // complete this part
    }
```

5. What do you understand by initialization of object?

Section B: Answer the following Questions:

(7x3=21 Marks)

1. Write a program to depict the function of unary operator overloading.
2. Write a program to implement binary operator overloading?
3. Differentiate between a constructor and a destructor.

Section C: Answer the following Questions:

(11x3=33 Marks)

1. What is function overloading? What are the advantages of function overloading? Write a program to calculate the area of circle, area square and area of triangle using function overloading.
2. What is operator overloading? Write a program to add two complex numbers using operator overloading.
3. Design a class Student in C++ with the following data members: roll number, name, and marks in three subjects.

**Implement:**

A default constructor to take input from the user and display student details, average marks and percentage.

